

TASK RECORD

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TASK: API, TYPES OF API AND API STATUS CODES

Task Description:

The task is to study and document the basic concepts of API, different types of APIs, and API status codes.

TASK 1: API

API stands for Application Programming Interface. API helps two different software applications communicate with each other. An API is a set of rules that allows one application to communicate with another application or service.

EXAMPLE:

The API acts as a connection between the app and the server and helps in sending and receiving the required data. I open a food delivery application and search for restaurants. The application can send a request to the server through an API. The server processes the request and sends the restaurant details back through the API.

STRUCTURE:

User



Application



API



Server / Database



Response



Application



User

An API is a set of rules that allows one application to communicate with another application or service.

TASK 2: HOW AN API WORKS

The basic working of API has four steps

1. Request
2. API receives the request
3. Server processes the request
4. Response

WORKING:

Request - The application sends a request to the API.

API receives the request - The API receives the request and passes it to the required server or service.

Server processes the request - The server checks the request and gets or changes the required data.

Response - The server sends a response back through the API.

EXAMPLE:

Student :

```
{  
  "name": "Arun",  
  "age": 20  
}
```

TASK 4: TYPES OF API

There are different types of API they are mainly classified into

1. REST
2. GraphQL
3. WEBSOCKET
4. GRPC
5. MQTT
6. SERVERLESS

1.REST API:

REST stands for Representational State Transfer. REST API is one of the commonly used for web applications. It uses HTTP methods to perform different operations. methods are GET, POST, PUT, PATCH, DELETE.

2. SOAP IS AN API:

SOAP stands for Simple Object Access Protocol. communication method that follows a fixed structure and uses XML to exchange data.

3. Graph QL:

Graph QL is a query language used with APIs. One important feature of Graph QL is that we can ask for only the information we need. we can request only the specific data we need from the server.

TASK3: WHAT IS AN API STATUS CODE?

An API status code is a number sent by the server in the response. It tells us what happened to the request. example, if the request was successful, the server may return:200 OK If the requested data cannot be found, it may return:404 Not Found Status codes are generally divided into five groups:

1 → Informational

2 → Success

3 → Redirection

4 → Client Error

5 → Server Error

IMPORTANT API STATUS CODES:

1.200 ok:

This means the request was successful.

2.201 Created:

This means a new resource was successfully created.

3.204 No content:

This means the request was successful, but there is no content to return in the response.

4.400Bad Request:

This means the request sent by the client is incorrect or missing required information.

5.401 Unauthorized :

This means authentication is required or the provided authentication information is not valid.

6.403 Forbidden:

This means the server understood the request but does not allow the client to access the requested resource.

7.404 Not Found:

This means the requested resource could not be found.

8.500 Internal Server Error:

This means something went wrong on the server while processing the request.

ADVANTAGES OF APIS:

1. They help different applications communicate with each other.
2. They make it easier to share data between applications.
3. They reduce the need to build everything from the beginning.
4. They are commonly used in websites and mobile applications.
5. They can connect applications with external services.

MY UNDERSTANDING:

From my understanding, an API acts like a connection between two applications. One application sends a request, the API passes the request to the required service, and then the response comes back from the API.

CONCLUSION:

From this topic, I understood that APIs help different applications communicate with each other and share information. Different methods like GET, POST, PUT, PATCH, and DELETE are used to perform different operations on data.